Introduction to database project:

PET ADOPTION SYSTEM

Team members:

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Name	Assigned responsibilities
مريم العيسى	ER model, ER model diagram, adopter table, 2 select queries, 2 aggregate functions
يسرى تامر	Relational schema, staff table, subquery
بتول حلاق	ER model scenario, relational schema (DB state), pet table, final report
غدير نفادي	Adoption application table, inner join, view

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SECTION 1: CONCEPTUAL AND LOGICAL DESIGN:

1.1 ER MODEL:

Entity	Attributes	Relationships
Pet	Pet_ID ,Name ,Species , Age ,Gender ,Health_Status	Each Pet can be linked to one or many Adoption_Applications
Adopter	Adopter_ID ,Name ,Phone , Address	One Adopter can submit many Adoption_Applications
Adoption_Application	Application_ID , Application_Date ,Pet_ID , Adopter_ID	Connects one or many Pets with one Adopter
Staff	Staff_ID ,Name ,Role ,Phone	Staff can review Adoption_Applications and monitor Pet health

1.2 ER MODEL SCENARIO:

The Pet Adoption System aims to organize and streamline the process of adopting pets.

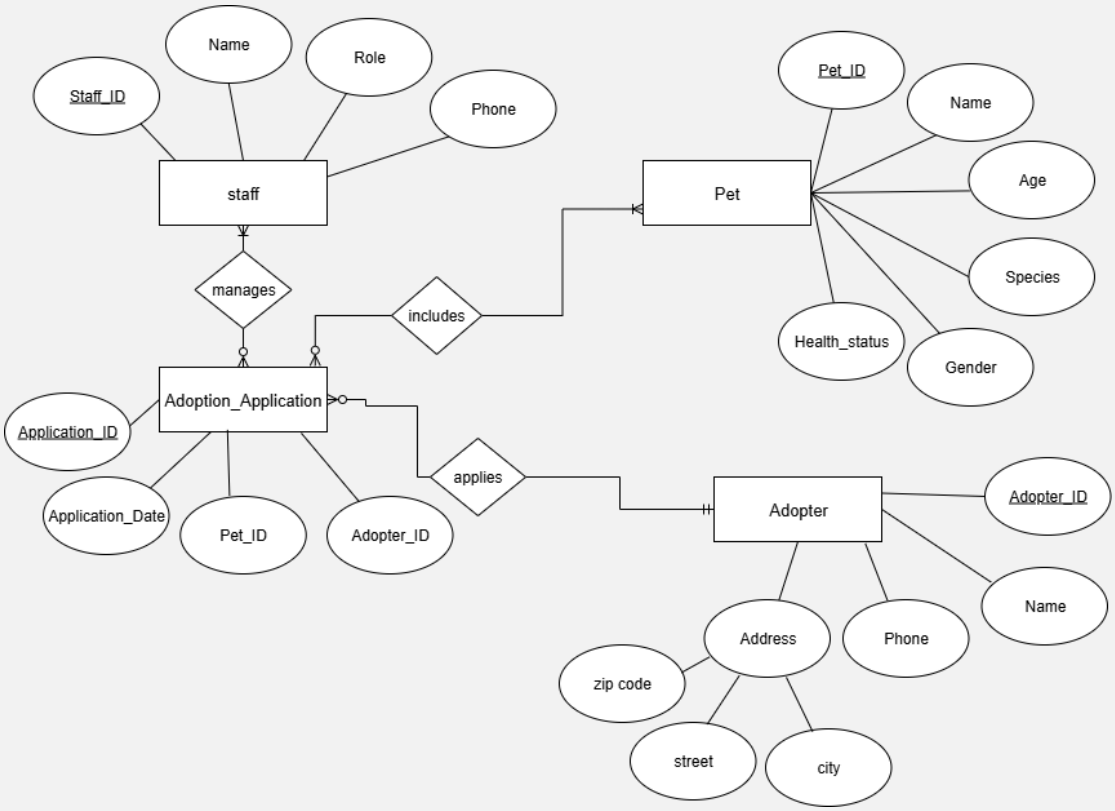
Pet: Each pet has a unique identifier (Pet_ID) with attributes such as name, species, age, gender, and health status. A pet can be linked to one or more adoption application.

Adopter: Each adopter is recorded with details such as name, phone number, and address. An adopter may submit multiple adoption applications for different pets.

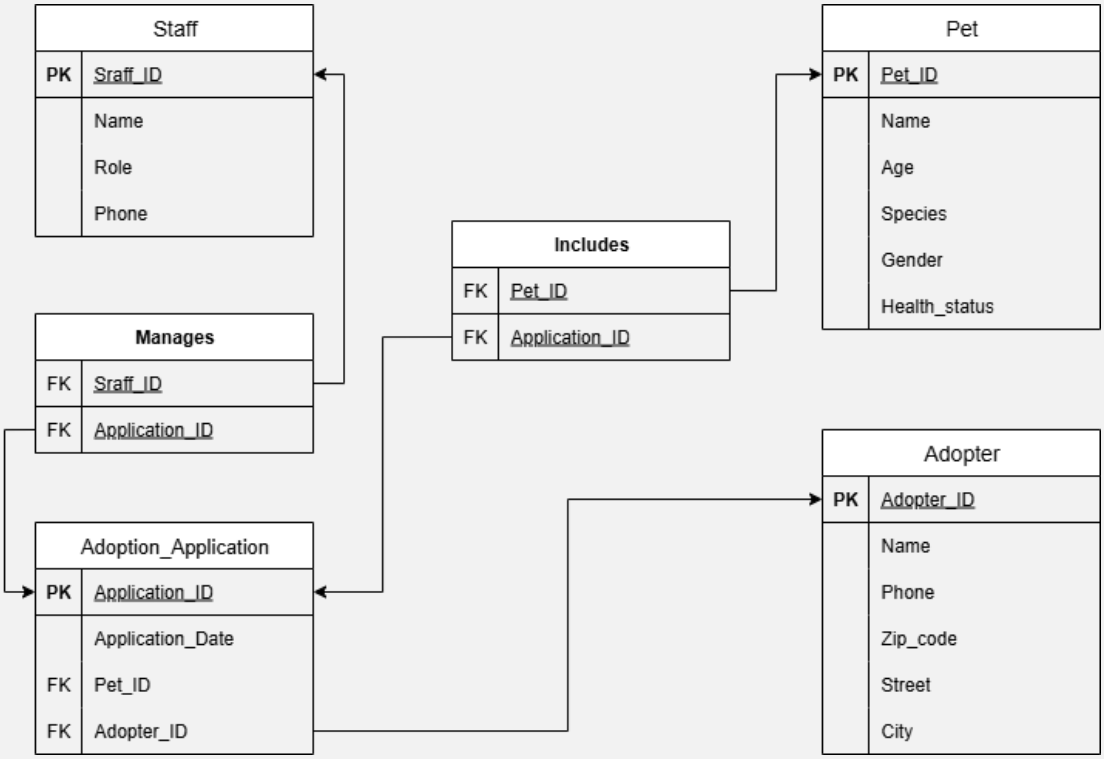
Adoption Application: This entity connects pets and adopters. It includes an application ID (Application_ID), date (Application_Date), and foreign keys for both the pet (Pet_ID) and adopter (Adopter_ID).

Staff: Staff members include veterinarians and administrators. Each staff member has a unique identifier (Staff_ID), name, role, and phone number. Staff are responsible for reviewing applications and monitoring pets' health.

1.3 ER MODEL DIAGRAM:



1.4 RELATIONAL SCHEMA:



Pet

Pet_ID	Name	Age	Species	Gender	Health status

Adopter

Adopter_ID	Name	Phone	Zip_code	Street	City

Adoption_Application

Application_ID	Application_Date	Pet_ID	Adopter_ID

Staff

Staff_ID	Name	Role	Phone

1.5 RELATIONAL SCHEMA (DB STATE):

Pet

Pet_ID	Name	Age	Species	Gender	Health status
101	Misho	2	Cat	Female	Healthy
102	Rocky	3	Dog	Male	Vaccinated
103	Lulu	1	Cat	Female	Needs Care
104	Max	4	Dog	Male	Healthy
105	Coco	5	Bird	Male	Injured Wing

Adopter

Adopter_ID	Name	Phone	Zip_code	Street	City
51	Sara Mohammed	0501234567	11461	King Fahd Rd	Riyadh
52	Ali Hassan	0539876543	21577	Prince Sultan St	Jeddah
53	Lina Abdullah	0592223344	32245	King Saud St	Dammam
54	Omar Khalid	0541122334	42313	Quba St	Madinah
55	Huda Ibrahim	0583344556	47911	King Abdulaziz Rd	Tabuk

Adoption_Application

Application_ID	Application_Date	Pet_ID	Adopter_ID
301	2025-09-01	101	51
302	2025-09-03	103	52
303	2025-09-05	102	53
304	2025-09-07	104	54
305	2025-09-09	105	55

Staff

Staff_ID	Name	Role	Phone
1	Dr. Ahmad Ali	Veterinarian	0507654321
2	Mona Saeed	Coordinator	0551239876
3	Khaled Omar	Admin	0561122443
4	Reem Salem	Veterinarian	0579988776
5	Youssef Nader	Volunteer	0586655443

SECTION 2: IMPLEMENTATION:

2.1 DATABASE CREATION:

Creating our database (Pet_Adoption) using **CREATE DATABASE** command:

```
create database Pet_Adoption;
```

Creating our tables with appropriate data types and constraints:

Adopter:

```
USE Pet_Adoption;
CREATE TABLE Adopter (
    Adopter_ID INT NOT NULL,
    Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(12) NOT NULL,
    Zip_code VARCHAR(10),
    Street VARCHAR(100),
    City VARCHAR(50),
    PRIMARY KEY (Adopter_ID)
    );
```

Pet:

```
USE Pet_Adoption;
    CREATE TABLE Pet(
    Pet_ID INT NOT NULL,
    Name VARCHAR(100),
    Species VARCHAR(100),
    Age INT(10),
    Gender VARCHAR(10),
    Health_Status VARCHAR(200),
    PRIMARY KEY (Pet_ID)
    );
```

Staff:

```
USE Pet_Adoption;
CREATE TABLE Staff(
    Staff_ID INT NOT NULL,
    Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(12) NOT NULL,
    Role VARCHAR(50),
    PRIMARY KEY (Staff_ID)
);
```

Adoption_Application:

```
USE Pet_Adoption;
CREATE TABLE Adoption_Application(
  Application_ID INT NOT NULL,
  Adopter_ID INT,
  Pet_ID INT,
  Application_Date DATE,
  PRIMARY KEY (Application_ID),
  FOREIGN KEY (Adopter_ID) REFERENCES Adopter(Adopter_ID),
  FOREIGN KEY (Pet_ID) REFERENCES Pet(Pet_ID)
);
```

Inserting values to our tables:

```
INSERT INTO Adopter (Adopter_ID, Name, Phone, Zip_code, Street, City)
VALUES
(51, 'Sara Mohammed', '0501234567', '11461', 'King Fahd Rd', 'Riyadh'),
(52, 'Ali Hassan', '0539876543', '21577', 'Prince Sultan St', 'Jeddah'),
(53, 'Lina Abdullah', '0592223344', '32245', 'King Saud St', 'Dammam'),
(54, 'Omar Khalid', '0541122334', '42313', 'Quba St', 'Madinah'),
(55, 'Huda Ibrahim', '0583344556', '47911', 'King Abdulaziz Rd', 'Tabuk');
```

```
INSERT INTO Pet (Pet_ID, Name, Age, Species, Gender, Health_Status)
VALUES
(101, 'Misho', 2, 'Cat',  'Female', 'Healthy'),
(102, 'Rocky', 3, 'Dog',  'Male',   'Vaccinated'),
(103, 'Lulu',  1, 'Cat',  'Female', 'Needs Care'),
(104, 'Max',   4, 'Dog',  'Male',   'Healthy'),
(105, 'Coco',  5, 'Bird', 'Male',   'Injured Wing');
```

```
INSERT INTO Staff (Staff_ID, Name, Role, Phone )
VALUES
(123, 'Ghadeer Sami', 'Admin', '0548775119'),
(321, 'Maryam Alessa', 'Data Analyst', '0555973709'),
(231, 'Batool Hossam', 'Veterinarian', '0567893708'),
(312, 'Yusra Tamer', 'Systems Programmer', '0541700270');
```

```
INSERT INTO Adoption_Application (Application_ID, Adopter_ID, Pet_ID, Application_Date)
VALUES
(9001, 51, 101, '2025-01-12'),
(9002, 52, 104, '2025-02-03'),
(9003, 53, 103, '2025-02-15'),
(9004, 54, 102, '2025-03-01'),
(9005, 55, 105, '2025-03-10');
```

2.2 APPLYING QUERIES:

Displaying pet ID, name and age from Pet table where pet age smaller then 3 using **SELECT** command:

```
USE Pet_Adoption;
SELECT Pet_ID, Name, Age
FROM Pet
WHERE Age < 3;
```

Output:

Pet_ID	Name	Age
101	Misho	2
103	Lulu	1

Displaying all column from Pet table where the pet' health status equal 'Healthy' using **SELECT** command:

```
USE Pet_Adoption;
SELECT *
FROM Pet
WHERE Health_Status = 'Healthy';
```

Output:

Pet_ID	Name	Species	Age	Gender	Health_Status
101	Misho	Cat	2	Female	Healthy
104	Max	Dog	4	Male	Healthy

Using **AVG() function** to calculate the average age of all pets in the *Pet* table.

```
USE Pet_Adoption;
SELECT AVG(Age) AS Average_Age
FROM Pet;
```

Output:

Average_Age
3.0000

Using **COUNT(*) function** to counts the total number of adoption applications recorded in the *Adoption_Application* table.

```
USE Pet_Adoption;
SELECT COUNT(*) AS Total_Applications
FROM Adoption_Application;
```

Output:

Total_Applications
5

Question: which adopter submitted an adoption application for the oldest pet?

```
/*
   This query returns the name of THE adopter who submitted adoption applications
   for the oldest pet in the shelter.
*/
SELECT Name
FROM Adopter
WHERE Adopter_ID IN (
    SELECT Adopter_ID
    FROM Adoption_Application
    WHERE Pet_ID IN (
        SELECT Pet_ID
        FROM Pet
        WHERE Age = (SELECT MAX(Age) FROM Pet)
    )
);
```

The inner subquery finds the **maximum pet age** in the *Pet* table. Using that age, the next query selects the **Pet_ID** of the oldest pet(s). The middle query retrieves the **Adopter_ID** of anyone who applied to adopt that pet. The outer query returns the **name of the adopter** who submitted that application.

Answer:

Name
Huda Ibrahim

Performing **INNER JOIN** between *Adoption_Application* table and *Pet* table by matching their shared key (Pet_ID). The result shows which pet is linked to each application, displaying adoption records with the corresponding pet names.

```
SELECT Adoption_Application.Application_ID, Pet.Name AS Pet_Name
FROM Adoption_Application
JOIN Pet
    ON Adoption_Application.Pet_ID = Pet.Pet_ID;
```

Application_ID	Pet_Name
9001	Misho
9002	Max
9003	Lulu
9004	Rocky
9005	Coco

Creating **view** named Vpet_Application that combines adoption application information with the corresponding pet names. Using INNER JOIN between the *Adoption_Application* and *Pet* tables to produce a virtual table containing the **application ID**, **application date**, and **pet name**.

```
CREATE VIEW Vpet_Application AS
SELECT Adoption_Application.Application_ID, Adoption_Application.Application_Date, Pet.Name AS Pet_Name
FROM Adoption_Application
JOIN Pet
ON Adoption_Application.Pet_ID = Pet.Pet_ID;

SELECT * FROM Vpet_Application;
```

Output:

Application_ID	Application_Date	Pet_Name
9001	2025-01-12	Misho
9002	2025-02-03	Max
9003	2025-02-15	Lulu
9004	2025-03-01	Rocky
9005	2025-03-10	Coco